An eighth-order inverse-response filter in a ten-site quantum walk

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Abstract

We construct a fixed, positive-coupling ten-site quantum walk with uniform eighth-order absolute inverse-response error. An interval contraction certifies simultaneous cancellation of the cubic, quintic and septic low-energy coefficients at dimensionless time 2π. A contour argument and a separate high-energy spectral-weight estimate prove an explicit bound for $0 < x \leq 1/16$, retaining the rapidly oscillating contribution. The corresponding postselected inverse has seventh-order conditional-state error under an explicit spectral-domain condition. Matched propagation examples report success probability and Hamiltonian norm cost. This resolves the four-extension existence question within the autonomous quantum-walk inversion model; the general chain-length conjecture remains open.

Keywords: continuous-time quantum walk; matrix inversion; spectral filter; interval arithmetic; postselection

1 Introduction

Weakly coupled quantum walks provide inverse-matrix responses through one evolution under a fixed Hamiltonian. In their preprint *Matrix Inversion by Quantum Walk* (arXiv:2508.06611v1, 2025), Kay and Tamon formulate an existence conjecture for chains with R extensions on either side of the matrix block. Their Sec. II.A, Table I and Appendix A give a ten-site, $R = 4$ design with absolute response error of order x^7 under a fixed low-energy spectrum. The desired unrestricted order is x^8. Their odd low-energy expansion and suppression of the high-energy endpoint weight supply the mechanism underlying the present construction.

We free one finite spectral eigenvalue and simultaneously cancel the cubic, quintic and septic coefficients. An exact root specified by an interval contraction fixes four positive couplings. We then bound the entire amplitude uniformly, including the high-energy oscillation. An ordinary Taylor expansion of the full Hamiltonian at $x = 0$ would miss this contribution and would not establish the claimed error order.

Related weak-coupling transfer results are developed in Kay and Tamon's *The strength of weak coupling* (arXiv:2512.08141v1, 2025), Theorem 4.1 and Secs. 4.2 and 6. Quantitative transfer estimates on graphs with an involution are also studied by Lippner and Shi [1]. These transfer settings differ from the fixed-time inverse filter considered here. Kang and Su [2], Sec. 4.4, Proposition 19 and Corollary 20 of their current preprint version, construct inverses through Hamiltonian singular-value transformation using repeated queries. That resource setting is distinct from one fixed autonomous chain. Our comparison concerns the nearest checked fixed-chain design; it makes no exhaustive priority, general-R, or complete algorithmic speedup claim.

2 Model and uniform response result

In an orthonormal single-excitation site basis, let

$$T_4(x;J)=\sum_{j=1}^{9} w_j\big(|j\rangle\langle j+1|+|j+1\rangle\langle j|\big), \qquad x>0, \tag{2.1}$$

where the open chain has zero diagonal and

$$(w_1,\dots,w_9)=(J_1,J_2,J_3,J_4,x^{-1},J_4,J_3,J_2,J_1). \tag{2.2}$$

Its full endpoint amplitude is

$$a_4(x;J,\tau)=\langle 10|e^{-\mathrm{i}\tau T_4(x;J)}|1\rangle. \tag{2.3}$$

All four couplings and the dimensionless time are fixed independently of x.

Theorem 1. *There are positive fixed couplings $J_1,\dots,J_4$ and a real $\ell\in(-1.078612,-1.078611)$ such that, at $\tau=2\pi$,*

$$\big|a_4(x;J,2\pi)-\mathrm{i}\ell x\big|\le 131072002\,x^8, \qquad 0<x\le\frac{1}{16}. \tag{2.4}$$

The couplings are specified exactly by the unique cancellation root in Sec. 3.2. For orientation, they are approximately

$$(J_1,J_2,J_3,J_4)\simeq(0.6803742065,\ 0.9435755016,\ 1.1381740446,\ 0.8283449422). \tag{2.5}$$

The displayed rounded values are not a substitute for the exact root.

The numerical constant in Eq. (2.4) is deliberately conservative. The conclusion is an existence statement in the finite Hermitian model, with a computer-assisted root certificate and analytic remainder estimates. It covers the $R=4$ case, without a construction for arbitrary R.

3 Reflection sectors and cancellation equations

3.1 Exact resolvent representation

Reflection about the central bond gives sectors spanned by $(|j\rangle\pm|11-j\rangle)/\sqrt{2}$, $j=1,\dots,5$. The two blocks $B_\pm(x)$ have off-diagonals $J_1,\dots,J_4$, and their only nonzero diagonal entry is the last, $\pm x^{-1}$. For $D=\mathrm{diag}(1,-1,1,-1,1)$, $B_-(x)=-DB_+(x)D$. Therefore

$$\begin{aligned} a_4(x;J,\tau)&=\frac{1}{2}\Big(\langle 1|e^{-\mathrm{i}\tau B_+(x)}|1\rangle-\langle 1|e^{-\mathrm{i}\tau B_-(x)}|1\rangle\Big)\\ &=-\mathrm{i}\sum_{r=1}^{5} W_r(x)\sin\big(\tau E_r(x)\big), \end{aligned} \tag{3.1}$$

where E_r are the eigenvalues of B_+ and W_r their first-site spectral weights. In particular, no extra factor of two enters the endpoint response.

Write $a=J_1^2$, $b=J_2^2$, $c=J_3^2$, $d=J_4^2$ and set

$$\begin{aligned} p(z)&=z^4-(a+b+c)z^2+ac, & q(z)&=zp(z)-dz(z^2-a-b),\\ n_0(z)&=z^3-(b+c)z, & n_1(z)&=zn_0(z)-d(z^2-b). \end{aligned} \tag{3.2}$$

Continuant recursion for the determinant and first-site minor gives

$$\det(zI-B_+)=-\frac{p(z)-xq(z)}{x}, \qquad G_{11}(z,x)=\frac{n_0(z)-xn_1(z)}{p(z)-xq(z)}. \tag{3.3}$$

The polynomial identity $n_0 q - n_1 p = -abcd$ yields

$$G_{11}(z,x) = \frac{n_0(z)}{p(z)} - abcd \sum_{n\geq 1} x^n \frac{q(z)^{n-1}}{p(z)^{n+1}}, \qquad |xq/p| < 1. \tag{3.4}$$

Let $a_{\text{low}} = \mathrm{i} f(x)$ denote the contribution from the four finite eigenvalues inside a fixed contour. The paired zero-x spectrum makes $f_0 = 0$. Contour integration of Eq. (3.4) gives

$$f(x) = \sum_{n\geq 1} f_n x^n, \qquad f_n = abcd\, S_n, \qquad S_n = \sum_{p(\rho)=0} \mathrm{Res}_{z=\rho} \frac{\sin(\tau z) q(z)^{n-1}}{p(z)^{n+1}}. \tag{3.5}$$

Since p is even and q is odd, the integrand is even for even n, and opposite-root residues cancel. Thus every even coefficient vanishes. Only the low-energy part is expanded here.

3.2 An exactly specified positive root

Fix $u = 1/2$ and $\tau = 2\pi$. Parametrize

$$\begin{aligned} p(z) &= (z^2 - u^2)(z^2 - v^2), \\ a &= \frac{u^2 v^2}{c}, \qquad b = u^2 + v^2 - c - \frac{u^2 v^2}{c}, \qquad s = a + b = u^2 + v^2 - c. \end{aligned} \tag{3.6}$$

The domain $u^2 < c < v^2$, $v > u$, $d > 0$ makes all four squared couplings positive: $bc = (c - u^2)(v^2 - c) > 0$. The three equations are

$$F(v, c, d) = (S_3, S_5, S_7) = 0. \tag{3.7}$$

They are real analytic on the box used below. Each residue is a finite polynomial-jet calculation. More explicitly, for $\rho \in \{u, v\}$ and the other positive root ω,

$$\begin{aligned} p(\rho + y) &= y h_\rho(y), \\ h_\rho(y) &= 2\rho(\rho^2 - \omega^2) + (5\rho^2 - \omega^2)y + 4\rho y^2 + y^3, \qquad (3.8) \\ S_n &= 2 \sum_{\rho = u, v} [y^n] \left\{ \sin\big(2\pi(\rho + y)\big) q(\rho + y)^{n-1} h_\rho(y)^{-n-1} \right\}, \qquad n \text{ odd}. \qquad (3.9) \end{aligned}$$

This makes the cancellation equations reproducible without fitting a singular full propagator.

Let X be the closed infinity-norm box of radius $r = 10^{-40}$ about the exact terminating decimal vector $y_0 = (v_0, c_0, d_0)$:

$$\begin{aligned} v_0 &= \texttt{1.5487685249240345145585943938889954395158348752808834652101731657} \\ c_0 &= \texttt{1.2954401557316677104527788216689492776181313419030944437678424473} \\ d_0 &= \texttt{0.6861553432828774826400267327924728455072128224835685906732202562 7} \end{aligned} \tag{3.10}$$

Using a fixed exact decimal preconditioner Y (Appendix A), define $\Phi(y) = y - YF(y)$. Elementary directed interval arithmetic at 80 decimal digits, with an analytically differentiated jet recurrence, encloses the complete Jacobian on X and gives

$$\begin{aligned} \sup_{y\in X} \|I - YDF(y)\|_\infty &< 8.0 \times 10^{-36}, \\ \|YF(y_0)\|_\infty &< 4.28 \times 10^{-65}, \\ \sup_{y\in X} \|\Phi(y) - y_0\|_\infty &< 4.28 \times 10^{-65} < r. \end{aligned} \tag{3.11}$$

The last line includes the Jacobian contribution times r, rather than only the center displacement. Thus Φ maps X strictly into itself and is a contraction. Banach's fixed-point theorem gives a

unique fixed point. The first inequality also makes $YDF(y)$ invertible; hence Y is invertible and the fixed point is an exact zero of F. Conversely every zero is a fixed point, establishing uniqueness within X.

At this zero, define $J = (\sqrt{a}, \sqrt{b}, \sqrt{c}, \sqrt{d})$ using Eq. (3.6), and $\ell = abcd\, S_1$. The same interval calculation gives positivity and

$$-1.078612 < \ell < -1.078611. \tag{3.12}$$

Consequently $f_3 = f_5 = f_7 = 0$ exactly, and $f(x) = \ell x + O(x^9)$. These cancellations hold at the enclosed zero; the terminating decimal center has small nonzero residuals.

The supplementary certificate retains all exact binary interval endpoints. The enclosure uses only elementary arithmetic, sine, cosine and π from `mpmath.iv` 1.3.0, with $\sin \pi = 0$ and $\cos \pi = -1$ inserted exactly at $u = 1/2$. Numerical root finding and inversion propose the center and Y; their numerical accuracy is not an existence premise. This is a computer-assisted proof relying on the interval implementation, without proof-assistant formalization. A separate calculation from implicit finite-eigenvalue series and cofactor weights reproduces the coefficients through order seven; that numerical agreement supports the implementation and does not replace the contraction inequalities.

4 Uniform control of the complete amplitude

4.1 The analytic low-energy remainder

The root box implies the coarse rational bounds

$$\frac{3}{2} < v < \frac{31}{20}, \quad 1.2954 < c < 1.2955, \quad 0 < d < 0.687, \quad 0 < s < 1.36, \quad 0 < abcd < 1. \tag{4.1}$$

On the circle $|z| = 2$,

$$\begin{aligned} |p(z)| &\geq (4 - u^2)(4 - v^2) > 5, \\ \left|\frac{q(z)}{p(z)}\right| &\leq 2 + \frac{2d(4+s)}{5} < 4. \end{aligned} \tag{4.2}$$

For $|x| \leq 1/16$, $|xq/p| < 1/4$. Rouché's theorem counts exactly four zeros of $p - xq$ inside this circle. The contour integral is analytic in x and Eq. (3.4) converges uniformly there, so coefficient extraction under the integral is justified.

Since $|\sin(2\pi z)| \leq e^{4\pi} < 300000$, the circle length divided by 2π is 2, and $abcd < 1$, Eq. (3.5) gives

$$|f_n| \leq 24000\, 4^{n-1}, \qquad n \geq 1. \tag{4.3}$$

The elementary exponential bound can itself be checked rationally: use $\pi < 22/7$, sum $\sum_{k=0}^{100} (88/7)^k/k!$, and bound the positive tail by its first term times $1/(1 - (88/7)/102)$. The resulting upper bound is less than 300000. After the odd cancellations and even parity zeros, the remainder starts at order nine. For real $0 < x \leq 1/16$,

$$\begin{aligned} |f(x) - \ell x| &\leq \frac{24000\, 4^8 x^9}{1 - 4x} \\ &\leq 2097152000\, x^9 \leq 131072000\, x^8. \end{aligned} \tag{4.4}$$

Including the zero even coefficients in the geometric majorant only enlarges this bound.

4.2 The oscillatory high-energy branch

For real positive x, $B_+(x)$ is Hermitian. Its largest eigenvalue Λ satisfies $\Lambda \geq 1/x$ by the last-site Rayleigh quotient. Since $1/x \geq 16$, this is the fifth eigenvalue outside the contour. Let η be the last component of its normalized eigenvector. Solving the first four component equations gives

$$\langle 1|\psi_{\text{high}}\rangle = \eta \frac{J_1 J_2 J_3 J_4}{p(\Lambda)}. \tag{4.5}$$

The numerator follows from the cross-chain cofactor of the four-site principal block. With $|\eta| \leq 1$, its first-site weight obeys

$$\begin{aligned} W_{\text{high}} &\leq \frac{abcd}{p(\Lambda)^2} \\ &\leq \frac{x^8}{\left[(1-u^2x^2)(1-v^2x^2)\right]^2} \leq 2x^8. \end{aligned} \tag{4.6}$$

The last inequality uses $u^2 < v^2 < 3$, $x^2 \leq 1/256$ and $(253/256)^4 > 1/2$. Equation (3.1) identifies its full endpoint contribution as

$$a_{\text{high}}(x) = -\mathrm{i} W_{\text{high}}(x) \sin\bigl(2\pi\Lambda(x)\bigr). \tag{4.7}$$

Thus $|a_{\text{high}}| \leq 2x^8$ for every x in the specified interval. No expansion or averaging of this phase has been used. Adding Eq. (4.4) proves Theorem 1.

For the specified witness, perturbation theory for the isolated eigenvalue near 1 of $xB_+(x)$ gives $\Lambda = x^{-1} + dx + O(x^3)$ and $W_{\text{high}} = abcd\, x^8(1 + O(x^2))$. Therefore

$$a_4(x) - \mathrm{i}\ell x = -\mathrm{i} abcd\, x^8 \sin\bigl(2\pi/x + O(x)\bigr) + O(x^9). \tag{4.8}$$

Sequences with opposite sine maxima show

$$\limsup_{x\to 0^+} \frac{|a_4(x) - \mathrm{i}\ell x|}{x^8} = abcd \simeq 0.3663437609. \tag{4.9}$$

This explains the attained order for this witness and the oscillatory error envelope. It is not a general-chain optimality theorem.

5 Postselected inverse, probability and cost

Let $A = U\Sigma V^\dagger$ be square and invertible, with dimensionless singular values in $[1/\kappa, 1]$, $\kappa \geq 1$, and $\|A\| = 1$. In units $\hbar = 1$, introduce a common energy scale E_0. A ten-block Hamiltonian has the four couplings $\gamma J_j E_0$ on each side, central block $E_0 A$ and its adjoint, and zero block diagonal. For each singular channel,

$$H_\lambda/E_0 = \gamma T_4(\gamma/\lambda; J), \qquad t = \frac{2\pi}{\gamma E_0}, \qquad 0 < \gamma \leq \frac{1}{16\kappa}. \tag{5.1}$$

Thus $x = \gamma/\lambda$ remains inside the theorem's domain simultaneously for the entire input spectrum. Sites before the central block carry left singular vectors and sites after it carry right singular vectors. Endpoint detection gives the Kraus block

$$K = V \operatorname{diag}\bigl(a_4(\gamma/\lambda_j)\bigr) U^\dagger, \qquad K_0 = \mathrm{i}\ell\gamma A^{-1}. \tag{5.2}$$

With $C = 131072002$,

$$\|K - K_0\| \leq \Delta, \qquad \Delta = C(\gamma\kappa)^8. \tag{5.3}$$

For normalized b, define $n = \|K_0 b\| = |\ell|\gamma\|A^{-1}b\|$. Triangle inequalities and the contraction property of K give

$$\max(0, n-\Delta)^2 \le p_{\mathrm{succ}} = \|Kb\|^2 \le \min(1, (n+\Delta)^2). \tag{5.4}$$

If $\Delta < n$, the successful branch is nonzero and

$$\left\| \frac{Kb}{\|Kb\|} - \frac{K_0 b}{n} \right\| \le \frac{2\Delta}{n} \le \frac{2C}{|\ell|}\gamma^7\kappa^8. \tag{5.5}$$

For the first inequality, add and subtract Kb/n and use $|\|Kb\| - n| \le \Delta$. The second uses $\|A^{-1}b\| \ge 1$. The conditional reference includes the global phase $\mathrm{i}\ell/|\ell|$. The error power loses one order on normalization: the absolute response order is eight, whereas this conditional-state bound is seventh order for fixed κ and input. Rectangular inverses require a specified singular support and are not claimed here.

The complete evolution is unitary. The selected map $\rho \mapsto K\rho K^\dagger$ is completely positive and trace nonincreasing, with probability in Eq. (5.4). The construction does not yield deterministic inversion. Couplings stay of order one and $\tau = 2\pi$ matches the source design. A row-sum estimate gives the coarse dimensionless cost bound

$$\|H\| t \le 2\pi \left(\gamma^{-1} + 2 \max_j J_j \right). \tag{5.6}$$

Unamplified attempts have expected repetition count $1/p_{\mathrm{succ}}$. State preparation, Hamiltonian simulation, amplification, coherence and control precision remain additional resources.

6 Propagation comparison and limitations

The fixed-spectrum reference keeps $u = 1/2$, $v = 3/2$ and numerically cancels S_3, S_5 at high precision. Its couplings reproduce the rounded $R = 4$ values in Kay and Tamon's Table I. Its response coefficient is about -0.9430116913, and its septic coefficient is about -0.2639054062. The new witness frees $v \simeq 1.5487685249$ and cancels S_7 as well. The largest coupling increases from about 0.9926 to 1.1382, while the linear response magnitude increases to about 1.0786. Neither time nor couplings diverge to obtain the higher order.

Table 1 compares complete ten-site propagations at $x = 4/401$, where $2\pi/x$ is a sine maximum. These are scalar channel diagnostics with identical x and τ. The prior reference is recomputed at high precision rather than taken from six-digit printed couplings. It is a precision-stabilized numerical comparison, not a separate interval-certified prior theorem.

Design	$\lvert a_4 - \mathrm{i}\ell x\rvert$	$\lvert a_4\rvert^2$	$\lVert T_4\rVert\tau$
Fixed-spectrum reference	2.61870×10^{-15}	8.84841×10^{-5}	629.92122
Certified-root design	3.27585×10^{-17}	1.15761×10^{-4}	629.93233

Table 1: Complete scalar-channel propagation at $x = 4/401$ and $\tau = 2\pi$. The fixed-spectrum reference is reconstructed from the $R = 4$ design in Kay and Tamon, *Matrix Inversion by Quantum Walk* (arXiv:2508.06611v1), Table I. Each error is measured relative to that design's own linear coefficient. Displayed digits are numerical diagnostics; the uniform bound follows from the root certificate and analytic estimates.

The calculations compare a direct ten-site representation with the five-site spectral sum and repeat at 60 and 90 decimal digits. Across six tested $x = 4/(4k+1)$ values, $k = 25, 50, 100, 200, 500, 1000$, the representation discrepancies are below 10^{-89} and the precision differences below 7×10^{-61}. Coupling derivatives and Duhamel's inequality bound the response

uncertainty from the 10^{-40} parameter box by 40×10^{-40} over these propagations. These diagnostics support the calculations; the interval root and uniform inequalities in Sec. 4, rather than the samples, prove Eq. (2.4). The leading scalar success-probability ratio is approximately $(1.0786116373/0.9430116913)^2 \simeq 1.31$.

The sample shows substantially smaller response error at nearly equal $\|H\|t$. It does not establish a globally optimized filter, a complete quantum algorithm advantage, or robustness to timing, coupling and hardware errors. The uniform constant is loose; practical design would require tighter constants and an implementation-error budget. Only the exact $R = 4$ existence slice is resolved. Existence and efficient discovery for general R remain outside the result. The nearest-primary-source comparison is bounded and does not establish exhaustive priority.

Computational methods and AI disclosure

The interval calculation uses `mpmath` 1.3.0. Symbolic determinant and cofactor checks use `SymPy` 1.14.0; the numerical replay uses arbitrary-precision Hermitian spectral propagation. Supplementary code and exact certificate data specify the recurrences, precision settings, parameter box and decimal preconditioner. OpenAI Codex language-model tools were used for literature screening, derivation and root-search assistance, verification-code generation, computational interpretation and manuscript drafting. A separate AI checking stage used a different coefficient representation; it is not external peer review. This disclosure covers generative scientific work as well as writing. Human scientific review and authorship confirmation are pending for this local draft; completed human verification is not asserted.

Statements and declarations

Data and code availability. The accompanying local supplement contains the root certificate, propagation comparison data and scripts, symbolic checks, environment specification and regeneration instructions. It contains no experimental dataset. Public archival deposition has not occurred.

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Author information and contributions. The displayed author information is supplied for this draft and requires confirmation. The named author's scientific review, contribution statement and approval of the final text remain pending.

Ethics approval and consent. Not applicable to this finite-dimensional theoretical and computational study.

A Exact interval certificate inputs

The fixed exact decimal preconditioner used in Eq. (3.11) is listed row by row below. Each entry is a terminating decimal, interpreted exactly before outward rounding. The entries and center were proposed numerically; the strict interval inequalities provide the certificate.

```
Y11 = 0.06120712115527357521389077942907453743786782507065933407216405752 8
Y12 = -0.14614944776142125586130034028698605531319229338275959795286583887
Y13 = 0.03338874446906369632384810861182049655926106231111215223456165983 9
Y21 = 0.09598899350046434332223632483543832535925062089859951137310929682 3
Y22 = -0.48302764580534128969674320644336240119688446294722354899867335999
Y23 = 0.20656203665781233635385480955993131085603466502519415183850412 84
Y31 = 0.34065699313566067153924085606325995484979187065078198062875093929
Y32 = -0.25729234664620376978066335496729766930071543693004756949961486848
Y33 = 0.14478866502255163895363617780142713794615098484178861690487505011
```

The stored certificate supplies the full interval Jacobian and the exact binary endpoints of $I - YDF(X)$, center corrections, image bounds, coupling-positivity intervals and linear response. The replay instructions evaluate the finite jets in Eq. (3.9) and differentiate them analytically. A second representation solves the four implicit eigenvalue series of $p(E) - xq(E) = 0$, reconstructs first-site weights as $(n_0(E) - xn_1(E))/(p'(E) - xq'(E))$, and composes their sine series. This separates implementation checks from the analytic treatment of the high-energy mode.

References

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